

Lab files

Thursday, July 10, 2025 1:02 PM

http://github.com/MalwareCube/SOC101/blob/main/course_files/02_Network_Security.zip

File password :

LCty2JmHQLB3uc9VxjeohENr

Network Security Theory

Sunday, June 29, 2025 1:37 PM

This a refresh I you don't know or have networking basics => jermmy IT lab CCNA in 60 days on YouTube

Network Layer Protocols

- **Internet Protocol (IP)**
 - IP addressing
 - IP routing
- **Transmission Control Protocol (TCP)**
 - Connection-oriented - reliable
 - Three-way handshake
 - Flow and congestion control
- **User Datagram Protocol (UDP)**
 - Connectionless
 - Lightweight, low-overhead - speed over reliability
 - VoIP, video streaming, audio streaming

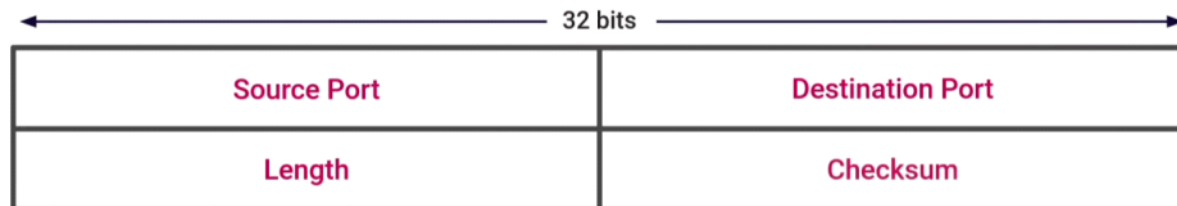


TCM

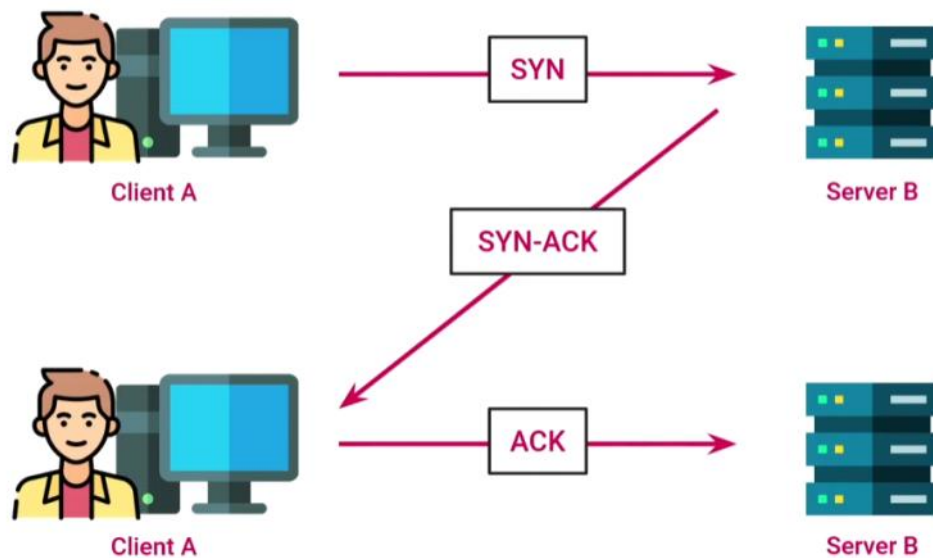
Internet Protocol Version 4 (IPv4) Header

32 bits				
Version	Header Length	Type of Service	Total Length	
Identification			Flags	Fragment Offset
Time To Live (TTL)		Protocol	Header Checksum	
Source Address				
Destination Address				
Options			Padding	

User Datagram Protocol (UDP) Header



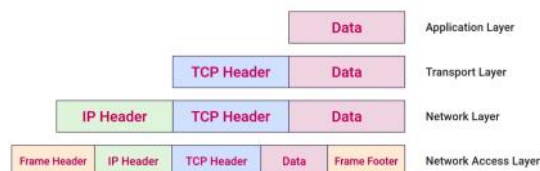
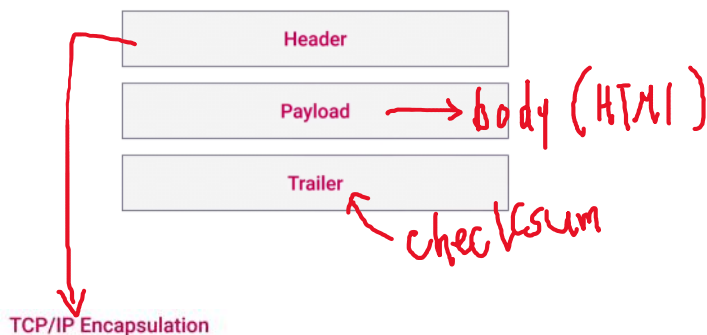
Three-Way Handshake



Common Ports

- **21 - File Transfer Protocol (FTP)**
 - Transfer files between systems
- **22 - Secure Shell (SSH)**
 - Often used for a secure remote command-line interface
- **23 - Telnet**
 - Unencrypted remote command-line interface
- **25 - Simple Mail Transfer Protocol (SMTP)**
 - Sending emails
- **53 - Domain Name System (DNS)**
 - Translating domain names into IP addresses
- **80 - HyperText Transfer Protocol (HTTP)**
 - Data communication over the web
- **110 - Post Office Protocol 3 (POP3)**
 - Retrieving emails from a remote server
- **135 - Microsoft Remote Procedure Call (RPC)**
 - Microsoft inter-process communication
- **139 - NetBIOS**
 - Windows file and printer sharing
- **143 - Internet Message Access Protocol (IMAP)**
 - Retrieving emails from a remote server
- **389 - Lightweight Directory Access Protocol (LDAP)**
 - Centralized directory and authentication service
- **443 - HyperText Transfer Protocol (HTTPS)**
 - HTTP with encrypted communications
- **445 - Server Message Block (SMB)**
 - Network file and printer sharing
- **3389 - Remote Desktop Protocol (RDP)**
 - Graphical interface to connect with a system
- **8080 - HyperText Transfer Protocol (HTTP)**

Packets



The PCAP is about seeing the routing info and the details

Packet Capture (PCAP)

Intercepting Packets

- Port mirroring (SPAN Ports)
- Inline network devices
- Network taps

Make a switch to copy traffic from port or more and sent to analysis
Capture packets and analyze directly like the IPS or IDS
Deceives connected to the network that can capture the traffic

PCAP Files

- Binary file format
- Header and raw payload
- Tcpdump, Wireshark, Tshark, etc.

Advantages

- Most detailed record
- Includes packet payload

Disadvantages

- Resource intensive
- Requires significant storage space
- Hard to scale

Date	Time	Call Type	Number Dialed	Duration	Cost	Transcript
2024-05-01	14:35	Local	(555) 123-4567	10	\$1.00	▶ "Hello. I'm just calli..."
2024-05-01	16:53	Local	(555) 987-6543	5	\$0.50	▶ "Hey Bill, do you thi..."
2024-05-02	09:12	Long Dist	(800) 555-1234	20	\$5.00	▶ "Is this the Krusty K..."
2024-05-02	18:06	Intl	+44 20 7946 0958	15	\$10.00	▶ "Hi there. You hav..."
2024-05-03	20:45	Local	(555) 555-5555	30	\$3.00	▶ "Congratulations! Y..."
2024-05-04	07:27	Long Dist	(888) 555-6789	25	\$6.25	▶ "You're late for work!"
2024-05-05	22:02	Intl	+81 3-1234-5678	10	\$7.50	▶ "Why do you keep c..."
2024-05-05	15:25	Local	(555) 222-3333	40	\$4.00	▶ "Hi there. My name i..."
2024-05-06	19:12	Local	(555) 444-4444	12	\$1.20	▶ "Are you free this w..."
2024-05-06	21:42	Long Dist	(900) 555-7890	18	\$4.50	▶ "Hi Bill. It's Mary. Do..."

Flow records is about normal and routing information from routers

Flow Records

Aggregated metadata

5-Tuple

- Source IP
- Source Port
- Destination IP
- Destination Port
- Transport Protocol

Advantages

- Efficient bandwidth and storage requirements
- High level pattern and anomaly detection
- Easier to scale

Disadvantages

- Lacks payload detail
- Doesn't give you the whole picture

Date	Time	Call Type	Number Dialed	Duration (min)	Cost
2024-05-01	14:35	Local	(555) 123-4567	10	\$1.00
2024-05-01	16:53	Local	(555) 987-6543	5	\$0.50
2024-05-02	09:12	Long Dist	(800) 555-1234	20	\$5.00
2024-05-02	18:06	Intl	+44 20 7946 0958	15	\$10.00
2024-05-03	20:45	Local	(555) 555-5555	30	\$3.00
2024-05-04	07:27	Long Dist	(888) 555-6789	25	\$6.25
2024-05-05	22:02	Intl	+81 3-1234-5678	10	\$7.50
2024-05-05	15:25	Local	(555) 222-3333	40	\$4.00
2024-05-06	19:12	Local	(555) 444-4444	12	\$1.20
2024-05-06	21:42	Long Dist	(900) 555-7890	18	\$4.50

Introduction to tcpdump

Monday, July 7, 2025 12:45 PM

Is a command line packet capturing tool

On windows we have windump

```
$ sudo apt-get install tcpdump
```

- In this tool we can do packet capture as well as we can make some scripts to capture as an event happens
- It captures packets directly from the network interface and it connects to the low-level OS

This as the filter or choosing interface options

```
sudo tcpdump [options] [expression]
```

This where we choose the thing the option will work on. As IP or network interface

To show the network interfaces you have :

```
m8a8a8y@m8a8a8y-VirtualBox: $ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd00::a00:27ff:fea4:86ed prefixlen 64 scopeid 0x0<global>
    inet6 fd00::1c0:350a:fc20:7ea prefixlen 64 scopeid 0x0<global>
    inet6 fe00::a00:27ff:fea4:86ed prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:a4:86:ed txqueuelen 1000 (Ethernet)
    RX packets 268 bytes 323948 (323.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 178 bytes 22396 (22.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 44 bytes 5484 (5.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 44 bytes 5484 (5.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Or

```
m8a8a8y@m8a8a8y-VirtualBox: $ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:a4:86:ed brd ff:ff:ff:ff:ff:ff
```

This command will show the interfaces that tcpdump can use:

```
m8a8a8y@m8a8a8y-VirtualBox: $ tcpdump -D
1.enp0s3 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.nflog (Linux netfilter log (NFLOG) interface) [none]
6.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
7.dbus-system (D-Bus system bus) [none]
8.dbus-session (D-Bus session bus) [none]
```

Using this command you can select the interface either the number or with the name (lo for 3)

```
m8a8a8y@m8a8a8y-VirtualBox: $ tcpdump -D
1.enp0s3 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.nflog (Linux netfilter log (NFLOG) interface) [none]
6.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
7.dbus-system (D-Bus system bus) [none]
8.dbus-session (D-Bus session bus) [none]
m8a8a8y@m8a8a8y-VirtualBox: $ sudo tcpdump -i 3
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on lo, link-type EN10MB (Ethernet), snapshot length 262144 bytes
13:10:00.854634 IP localhost.51900 > _localdnsstub.domain: 9629+ AAAA? api.snapcraft.io. (34)
13:10:00.854791 IP localhost.57218 > _localdnsstub.domain: 56556+ A? api.snapcraft.io. (34)
```

```

m8a8a8y@m8a8a8y-VirtualBox:~$ tcpdump -D
1.enp0s3 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.nflog (Linux netfilter log (NFLOG) interface) [none]
6.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
7.dbus-system (D-Bus system bus) [none]
8.dbus-session (D-Bus session bus) [none]
m8a8a8y@m8a8a8y-VirtualBox:~$ sudo tcpdump -i 3
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on lo, link-type EN10MB (Ethernet), snapshot length 262144 bytes
13:10:00.854634 IP localhost.51900 > _localdnsstub.domain: 9629+ AAAA? api.snapcraft.io. (34)
13:10:00.854791 IP localhost.57218 > _localdnsstub.domain: 56556+ A? api.snapcraft.io. (34)
13:10:00.870621 IP _localdnsstub.domain > localhost.51900: 9629 4/3/0 AAAA 2620:2d:4000:1010::2e6, AAAA 2620:2d:4000:1010::42, AAAA 2620:2d:4000:1010::117 (213)
13:10:00.872820 IP _localdnsstub.domain > localhost.57218: 56556 4/3/0 A 185.125.188.58, A 185.125.188.57, A 185.125.188.54, A 185.125.188.59 (165)
13:10:00.893136 IP localhost.56598 > _localdnsstub.domain: 64528+ [1au] PTR? 53.0.0.127.in-addr.arpa. (52)
13:10:00.893661 IP _localdnsstub.domain > localhost.56598: 64528*$ 1/0/1 PTR _localdnsstub. (79)

```

Here we used net cat to make a listening section and on a port and another terminal listen contact that port and tcpdump see the packets

The image shows three terminal windows. The top window is a netcat listener on port 4321. The middle window is a netcat client connecting to 127.0.0.1 on port 4321. The bottom window shows tcpdump output capturing the connection. A red box highlights the connection details, and a blue circle highlights the packet length of 6. Annotations explain the netcat command flags and the tcpdump output.

```

m8a8a8y@m8a8a8y-VirtualBox:~$ sudo tcpdump -i 3
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on lo, link-type EN10MB (Ethernet), snapshot length 262144 bytes
13:16:54.918639 IP localhost.42700 > _localdnsstub.domain: 32100+ [1au] PTR? 0.0.0.0.in-addr.arpa. (49)
13:16:54.918851 IP _localdnsstub.domain > localhost.42700: 32100 NXDomain*$ 0/0/1 (49)
13:16:55.000261 IP _localdnsstub.domain > _localdnsstub.domain: 43009+ [1au] PTR? 53.0.0.127.in-addr.arpa. (52)
13:16:55.000552 IP _localdnsstub.domain > localhost.48624: 43009*$ 1/0/1 PTR _localdnsstub. (79)
13:16:57.277674 IP localhost.53796 > localhost.4321: Flags [S], seq 2930519626, win 65495, options [mss 65495,sackOK,TS val 26669280 ecr 0,nop,wscale 7], length 0
13:16:57.277691 IP localhost.4321 > localhost.53796: Flags [S.], seq 686879756, ack 2930519627, win 65483, options [mss 65495,sackOK,TS val 26669280 ecr 26669280,nop,wscale 7], length 0
13:16:57.277705 IP localhost.53796 > localhost.4321: Flags [.] , ack 1, win 512, options [nop,nop,TS val 26669280 ecr 26669280], length 0
13:17:03.461624 IP localhost.mdns > mdns.mcast.net.mdns: 0 [ZQ] PTR (QM)? _lpp._tcp.local. PTR (QM)? _lpps._tcp.local. (45)
13:17:03.565402 IP localhost.37002 > _localdnsstub.domain: 65439+ [1au] PTR? 251.0.0.224.in-addr.arpa. (53)
13:17:03.582309 IP _localdnsstub.domain > localhost.37002: 65439 1/0/1 PTR mdns.mcast.net. (81)
13:17:09.052765 IP localhost.4321 > localhost.53796: Flags [P.], seq 1:7, ack 1, win 512, options [nop,nop,TS val 26681055 ecr 26669280], length 6
13:17:09.052852 IP localhost.53796 > localhost.4321: Flags [.] , ack 7, win 512, options [nop,nop,TS val 26681055 ecr 26681055], length 0

```

when the connection was received tcpdump so shows Sin [S] Sin-Ack [S.]
Ack [.]

when we sent hello we saw the push-Ack [P.] Ack [.]

If you saw the blue circle this the length of the packet this means there is somethings sent and next we will see how to view it

We restarted the listening on tcpdump but we added the -X (will print out the hex of message and the ASCII of it to be read format
(if we used -A we are going to see only ASCII no hex)

```
m8a8a8y@m8a8a8y-VirtualBox:~$ sudo tcpdump -i lo -X
[sudo] password for m8a8a8y:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on lo, link-type EN10MB (Ethernet), snapshot length 262144 bytes
13:36:03.069361 IP localhost.4321 > localhost.53796: Flags [P.], seq 686879763:686879769, ack 2938519627, win 512, options [nop,nop,TS val 27815071 ecr 26681055], length 6
    0x0000: 4500 003a 806d 4000 4006 bc4e 7f00 0001  E...m0.0..N....
    0x0010: 7f00 0001 10e1 d224 28f0 f413 aeac 2e4b  .....$(.....K
    0x0020: 8018 0200 fe2e 0000 0101 080a 01a8 6c9f  .....l.
    0x0030: 0197 1edf 6865 6c6c 6f0a                ..hello.
13:36:03.069583 IP localhost.53796 > localhost.4321: Flags [.], ack 0, win 512, options [nop,nop,TS val 27815071 ecr 27815071], length 0
    0x0000: 4500 0034 c56c 4000 4006 7755 7f00 0001  E..4.l0.0.wU....
    0x0010: 7f00 0001 d224 10e1 aeac 2e4b 28f0 f419  .....$(.....K(
    0x0020: 8010 0200 fe28 0000 0101 080a 01a8 6c9f  .....(.....l.
    0x0030: 01a8 6c9f                                ..l.
```

m8a8a8y@m8a8a8y-VirtualBox:~\$ nc -lvp 4321

Listening on 0.0.0.0 4321
Connection received on localhost 53796
hello
hello

m8a8a8y@m8a8a8y-VirtualBox:~\$ nc 127.0.0.1 4321

hello
hello
[]

tcpdump: Capturing Network Traffic

Tuesday, July 8, 2025 9:23 AM

- By default tcpdump translate ip addresses into there name and port name to remove it add -n

```
9277, ack 799, win 31970, length 506
20:32:52.564450 IP SOC101-ubuntu.57396 > 190.92.188.190.static.a2webhosting.com.https:
  win 0, length 0
^C
180 packets captured
180 packets received by filter
0 packets dropped by kernel
tcm@SOC101-ubuntu:~$ sudo tcpdump -i enp0s3 -n
```

- Host : will only filter if the host name is same as give here its tesla.com

```
m8a8a8y@m8a8a8y-VirtualBox:~$ sudo tcpdump -i enp0s3 -n host tesla.com
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
09:32:45.227069 IP 10.0.2.15.57606 > 2.18.51.207.443: Flags [S], seq 3395514882, win 64240, options [mss 1460,sackOK,TS val 810821854 ec
r 0,nop,wscale 7], length 0
09:32:45.245566 IP 10.0.2.15.57610 > 2.18.51.207.443: Flags [S], seq 3913323823, win 64240, options [mss 1460,sackOK,TS val 810821873 ec
r 0,nop,wscale 7], length 0
09:32:45.443614 IP 2.18.51.207.443 > 10.0.2.15.57606: Flags [S.], seq 47168001, ack 3395514883, win 65535, options [mss 1460], length 0
09:32:45.443718 IP 10.0.2.15.57606 > 2.18.51.207.443: Flags [.], ack 1, win 64240, length 0
09:32:45.444918 IP 10.0.2.15.57606 > 2.18.51.207.443: Flags [P.], seq 1:1895, ack 1, win 64240, length 1894
09:32:45.446076 IP 2.18.51.207.443 > 10.0.2.15.57606: Flags [.], ack 1461, win 65535, length 0
09:32:45.446078 IP 2.18.51.207.443 > 10.0.2.15.57606: Flags [.], ack 1895, win 65535, length 0
09:32:45.463602 IP 2.18.51.207.443 > 10.0.2.15.57610: Flags [S.], seq 47232001, ack 3913323824, win 65535, options [mss 1460], length 0
09:32:45.463647 IP 10.0.2.15.57610 > 2.18.51.207.443: Flags [.], ack 1, win 64240, length 0
09:32:45.465199 IP 10.0.2.15.57610 > 2.18.51.207.443: Flags [P.], seq 1:1895, ack 1, win 64240, length 1894
09:32:45.466154 IP 2.18.51.207.443 > 10.0.2.15.57610: Flags [.], ack 1461, win 65535, length 0
09:32:45.466155 IP 2.18.51.207.443 > 10.0.2.15.57610: Flags [.], ack 1895, win 65535, length 0
```

- Src & dst : are used to filter on source ip or destination ip

```
$ sudo tcpdump -i enp0s3 -n dst 142.251.33.174
```

- Net : is used to filter on specific subnet

```
$ sudo tcpdump -i enp0s3 -n net 192.168.1.0/24
```

```
$ sudo tcpdump -i enp0s3 -n src net 192.168.1.0/24
```

```
$ sudo tcpdump -i enp0s3 -n dst net 192.168.1.0/24
```

- Port will filter on a specific port

```
m8a8a8y@m8a8a8y-VirtualBox:~$ sudo tcpdump -i enp0s3 -n port 443
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
09:40:32.390241 IP 10.0.2.15.42688 > 216.58.211.206.443: Flags [S], seq 3815115952, win 64240, options [mss 1460,sackOK,TS val 274177180
4 ecr 0,nop,wscale 7], length 0
09:40:32.412303 IP 10.0.2.15.42704 > 216.58.211.206.443: Flags [S], seq 4207187704, win 64240, options [mss 1460,sackOK,TS val 274177182
7 ecr 0,nop,wscale 7], length 0
09:40:32.447638 IP 216.58.211.206.443 > 10.0.2.15.42688: Flags [S.], seq 108416001, ack 3815115953, win 65535, options [mss 1460], lengt
h 0
09:40:32.447707 IP 10.0.2.15.42688 > 216.58.211.206.443: Flags [.], ack 1, win 64240, length 0
09:40:32.448724 IP 10.0.2.15.42688 > 216.58.211.206.443: Flags [P.], seq 1:1900, ack 1, win 64240, length 1899
09:40:32.449550 IP 216.58.211.206.443 > 10.0.2.15.42688: Flags [.], ack 1461, win 65535, length 0
09:40:32.449551 IP 216.58.211.206.443 > 10.0.2.15.42688: Flags [.], ack 1900, win 65535, length 0
09:40:32.469577 IP 216.58.211.206.443 > 10.0.2.15.42704: Flags [S.], seq 108480001, ack 4207187705, win 65535, options [mss 1460], lengt
```

Also can specify either if wanted as source port or distention port

```
$ sudo tcpdump -i enp0s3 -n dst port 211
```

- Logical operators : and - not - or

```
$ sudo tcpdump -i enp0s3 -n src 10.0.2.10 and dst port 3389
```

- Single quotes : are used to make it easier to combine big complex filters also using the brackets

```
sudo tcpdump -i enp0s3 'src 192.168.1.1 and dst 192.168.1.2 and not ( port 21 or 80 )'
```

- Protocols : we can filter using the protocol name ip tcp udp icmp arp Etc

```
sudo tcpdump -i enp0s3 -n icmp
```

- Manual : see for more filters or usage

```
$ man tcpdump
```

- -w + path and name : to save or write the pcap file

```
$ sudo tcpdump -i enp0s3 -n host youtube.com -w Desktop/test.pcap
```

- -r + path and name : to read the pcap file

```
$ sudo tcpdump -r Desktop/test.pcap
```

- We can also make a filter on a saved pcap

```
m8a8a8y@m8a8a8y-VirtualBox:~$ sudo tcpdump -r Desktop/test.pcap port 443
reading from file Desktop/test.pcap, link-type EN10MB (Ethernet), snapshot length 262144
10:05:46.588271 IP m8a8a8y-VirtualBox.53616 > dub08s02-in-f78.1e100.net.https: Flags [S], seq 3386377428, win 64240, options [mss 1460,sa
ackOK,TS val 3038340617 ecr 0,nop,wscale 7], length 0
10:05:46.594952 IP m8a8a8y-VirtualBox.53622 > dub08s02-in-f78.1e100.net.https: Flags [S], seq 877573133, win 64240, options [mss 1460,sa
ckOK,TS val 3038340623 ecr 0,nop,wscale 7], length 0
10:05:46.644521 IP dub08s02-in-f78.1e100.net.https > m8a8a8y-VirtualBox.53616: Flags [S.], seq 305984001, ack 3386377429, win 65535, opti
ons [mss 1460], length 0
```

tcpdump: Analyzing Network Traffic

Tuesday, July 8, 2025 10:21 AM

- count : number of packets in a pcap

```
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop$ sudo tcpdump -r test.pcap --count
reading from file test.pcap, link-type EN10MB (Ethernet), snapshot length 262144
108 packets
```

- c num : will return the number of packets

```
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop$ sudo tcpdump -r test.pcap -c 2
reading from file test.pcap, link-type EN10MB (Ethernet), snapshot length 262144
10:05:46.588271 IP m8a8a8y-VirtualBox.53616 > dub88s02.in-f78.1e100.net.https: Flags [S], seq 3386377428, win 64248, options [mss 1460,sackOK,TS val 3038340617 ecr 0,nop,wscale 7], length 0
10:05:46.594952 IP m8a8a8y-VirtualBox.53622 > dub88s02.in-f78.1e100.net.https: Flags [S], seq 877573133, win 64248, options [mss 1460,sackOK,TS val 3038340623 ecr 0,nop,wscale 7], length 0
```

- t : will remove the time

But -tt and -tt -tttt all are format of time can be used

```
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop$ sudo tcpdump -r test.pcap -t
reading from file test.pcap, link-type EN10MB (Ethernet), snapshot length 262144
IP m8a8a8y-VirtualBox.53616 > nrs09s08.in-f14.1e100.net.https: Flags [S], seq 3386377428
IP m8a8a8y-VirtualBox.53622 > nrs09s08.in-f14.1e100.net.https: Flags [S], seq 877573133
IP nrs09s08.in-f14.1e100.net.https > m8a8a8y-VirtualBox.53616: Flags [S.], seq 305904001
IP m8a8a8y-VirtualBox.53616 > nrs09s08.in-f14.1e100.net.https: Flags [.], ack 1, win 6424
IP m8a8a8y-VirtualBox.53616 > nrs09s08.in-f14.1e100.net.https: Flags [P.], seq 1:1897, ad
IP nrs09s08.in-f14.1e100.net.https > m8a8a8y-VirtualBox.53616: Flags [.], ack 1461, win 6
IP nrs09s08.in-f14.1e100.net.https > m8a8a8y-VirtualBox.53616: Flags [.], ack 1897, win 6
IP nrs09s08.in-f14.1e100.net.https > m8a8a8y-VirtualBox.53622: Flags [S.], seq 306048001
IP m8a8a8y-VirtualBox.53622 > nrs09s08.in-f14.1e100.net.https: Flags [.], seq 1, win 6424
IP m8a8a8y-VirtualBox.53622 > nrs09s08.in-f14.1e100.net.https: Flags [P.], seq 1:1897, ad
```

- Practical analysis : lets get some hands dirty

```
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop/02_Network_Security/01_tcpdump/PCAPs$ ls
2021-09-14.pcap 2024-04-18.pcap
```

Look for :

- ☐ the conversation number of packets
- ☐ See the most common IPs are talking
- ☐ See the port or the protocol being used
- ☐ Try to dig deeper example (if its http try to look for get or post request or any file upload)

```
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop/02_Network_Security/01_tcpdump/PCAPs$ sudo tcpdump -r 2021-09-14.pcap --count
reading from file 2021-09-14.pcap, link-type EN10MB (Ethernet), snapshot length 262144
3679 packets
```

Most of the communication is between these 2 and its using http
IP 10.0.0.168.49679 > 103.232.55.148.http:

We are trying to see if there is any file upload or download or website that we can see

```
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop/02_Network_Security/01_tcpdump/PCAPs$ sudo tcpdump -r 2021-09-14.pcap host 10.0.0.168 and port 80
| grep -E "GET|POST"
```

```
6maxwidth=32768&rowheight=20&sectionHeight=160&FORM=IES502&market=en-US HTTP/1.1
17:35:31.035477 IP 10.0.0.168.49722 > 13.107.5.80.http: Flags [P.], seq 4294:4956, ack 5446, win 32768, length 662: HTTP: GET /qsnl.aspx?query=http3AN2FK2F103.232.55.148N2FserviceN2F.audlod
p&maxwidth=32768&rowheight=20&sectionHeight=160&FORM=IES502&market=en-US HTTP/1.1
17:35:31.201164 IP 10.0.0.168.49722 > 13.107.5.80.http: Flags [P.], seq 4956:5619, ack 6093, win 32687, length 663: HTTP: GET /qsnl.aspx?query=http3AN2FK2F103.232.55.148N2FserviceN2F.audlod
p&maxwidth=32768&rowheight=20&sectionHeight=160&FORM=IES502&market=en-US HTTP/1.1
17:35:31.407901 IP 10.0.0.168.49722 > 13.107.5.80.http: Flags [P.], seq 5619:6283, ack 6741, win 32686, length 664: HTTP: GET /qsnl.aspx?query=http3AN2FK2F103.232.55.148N2FserviceN2F.audlod
p&maxwidth=32768&rowheight=20&sectionHeight=160&FORM=IES502&market=en-US HTTP/1.1
17:35:31.703190 IP 10.0.0.168.49722 > 13.107.5.80.http: Flags [P.], seq 6283:6948, ack 7390, win 32768, length 665: HTTP: GET /qsnl.aspx?query=http3AN2FK2F103.232.55.148N2FserviceN2F.audlod
p&maxwidth=32768&rowheight=20&sectionHeight=160&FORM=IES502&market=en-US HTTP/1.1
17:35:31.840413 IP 10.0.0.168.49722 > 13.107.5.80.http: Flags [P.], seq 6948:7614, ack 8039, win 32686, length 666: HTTP: GET /qsnl.aspx?query=http3AN2FK2F103.232.55.148N2FserviceN2F.audlod
p&maxwidth=32768&rowheight=20&sectionHeight=160&FORM=IES502&market=en-US HTTP/1.1
17:35:32.552578 IP 10.0.0.168.49724 > 103.232.55.148.http: Flags [P.], seq 1:282, ack 1, win 32768, length 281: HTTP: GET /servlce/.audlod.exe HTTP/1.1
17:36:44.610317 IP 10.0.0.168.49745 > geschken.on3cx.de.http: Flags [P.], seq 1:252, ack 1, win 256, length 251: HTTP: POST /-element/page.php?id=484 HTTP/1.0
17:36:45.103946 IP 10.0.0.168.49744 > geschken.on3cx.de.http: Flags [P.], seq 1:252, ack 1, win 256, length 251: HTTP: POST /-element/page.php?id=484 HTTP/1.0
17:36:45.679415 IP 10.0.0.168.49745 > geschken.on3cx.de.http: Flags [P.], seq 1:252, ack 1, win 256, length 251: HTTP: POST /-element/page.php?id=484 HTTP/1.0
17:36:55.102390 IP 10.0.0.168.49747 > 93.184.220.29.http: Flags [P.], seq 1:241, ack 1, win 256, length 248: HTTP: GET /MFEWtZBNMEsw5TA2BgUrDgMCgUa8BQ50otxN2Fh0ZtLk2Bz85IP17wEWvdIQUTL3UIB
LV5uBu5q52F6K2BrK570VXj2kCEA177eL5ppmeL17jGdvdL0K3D HTTP/1.0
```

This file looks weird because its has (.) in the path this means a hidden file and if we look about the name of the file

It look a file form Microsoft



But its using http and is the IP its taken from for Microsoft ?
Will no

Quick Stats

IP Location	Viet Nam Ho Noi Viettelserver Services Technology Company Limited
ASN	AS61737 VIETSERVER-AS-VN VIETSERVER SERVICES TECHNOLOGY COMPANY LIMITED, VN (registered Aug 25, 2014)
Whois Server	whois.apnic.net
IP Address	103.232.55.148

🔍 Abuse contact for '103.232.52.0 - 103.232.55.255' is - hs-charge@nmc.vn

```
inetnum: 103.232.52.0 - 103.232.55.255
netname: VIETSERVER-AS-VN
descr: VIETSERVER SERVICES TECHNOLOGY COMPANY LIMITED
descr: Xu Khuê, Chu Phan ward, Me Linh district, Ha Noi city
country: VN
tech-c: NNA52-AP
mnt-by: VIET-AP
status: ALLOCATED PORTABLE
remarks: NNA52-AP
remarks: 2013-09-26: NNA52-AP-007
```

```
msf5@msf5kali:~$ sudo tcpdump -r 2021-09-14.pcap | grep "audiog.exe"
[sudo] password for msf5:
reading from file 2021-09-14.pcap, link type EN10MB (Ethernet), snapshot length 262144
17:35:31.840413 IP 10.0.0.160.4972 > 13.107.5.80: http: Flags [P], seq 694047614, ack 8039, win 32686, length 666: HTTP: GET /gnsl.aspx?query=http3XANZ2F2F183.232.55.148X2Fsvrloc&2F.audiog.exe&maxwidth=32765&rowheight=20&sectionHeight=160&FORM=TE5502&mark=en-US HTTP/1.1
17:35:32.552578 IP 10.0.0.160.4974 > 103.232.55.148: http: seq 1282, ack 1, win 32678, length 281: HTTP: GET /service/audiog.exe HTTP/1.1
```

Less is to make it easy to scroll

[illegible]

This url looks suspicious (don't forget to add defang)

[Join our Community](#) and enjoy additional community insights and crowd-sourced detections, plus an API key to [automate checks](#).

Security vendors' analysis		Do you want to automate checks?
BK Defender	Malware	C&PF Malicious
G Data	Malware	Webroot Malicious
Heslo	Clean	Arcabit Clean

Quick State	
IP Location	United States Redmond Microsoft Corporation
ASN	AS8068 MICROSOFT-CORP-MSN-AS-BLOCK, US [registered Mar 31, 1997]
Whois Server	whois.arin.net
IP Address	13.107.5.80
Reverse IP	1 website uses this address.

tcpdump Challenge 1

Thursday, July 10, 2025

1:01 PM

Challenge URL

- <https://challenges.malwarecube.com/#/c/bb2d4b38-3f4f-4639-8b09-ef17bb229b33>

Introduction to Wireshark

Wednesday, July 9, 2025 9:23 AM

Wireshark: Capture and Display Filters

Wednesday, July 9, 2025 10:26 AM

Capture filter is used to make less packet and traffic capture -----> for more info look at <https://wiki.wireshark.org/CaptureFilters>

Welcome to Wireshark

Capture

...using this filter: ☒ host google.com and (port 80)

- enp0s3
- any
- Loopback: lo
- bluetooth-monitor
- nflag
- nqueue
- dbus-system
- dbus-session
- Cisco remote capture: ciscodump
- DisplayPort AUX channel monitor capture: dpauxmon
- Random packet generator: randpkt
- systemd Journal Export: sdjournal
- SSH remote capture: sshdump
- UDP Listener remote capture: udpdump
- Wi-Fi remote capture: wifidump

Display filter is filtering the packet been already captured -----> for more info look at <https://wiki.wireshark.org/DisplayFilters>

No.	Time	Source	Destination	Protocol	Length	Info
643	11.998661934	10.0.2.15	8.8.8.8	TCP	74	44164 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519269544 TSecr=0 WS=128
644	12.004333547	10.0.2.15	8.8.8.8	TCP	74	44174 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519269550 TSecr=0 WS=128
645	12.257475204	10.0.2.15	8.8.8.8	TCP	74	44180 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519269803 TSecr=0 WS=128
646	13.014456691	10.0.2.15	8.8.8.8	TCP	74	[TCP Retransmission] 44174 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519270560 TSecr=0 WS=128
647	13.014624191	10.0.2.15	8.8.8.8	TCP	74	[TCP Retransmission] 44164 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519270560 TSecr=0 WS=128
648	13.270263023	10.0.2.15	8.8.8.8	TCP	74	[TCP Retransmission] 44180 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519270816 TSecr=0 WS=128
649	14.037893638	10.0.2.15	8.8.8.8	TCP	74	[TCP Retransmission] 44164 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519271583 TSecr=0 WS=128
650	14.037251826	10.0.2.15	8.8.8.8	TCP	74	[TCP Retransmission] 44174 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519271583 TSecr=0 WS=128
651	14.294180179	10.0.2.15	8.8.8.8	TCP	74	[TCP Retransmission] 44180 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519271840 TSecr=0 WS=128
652	15.002093658	10.0.2.15	8.8.8.8	TCP	74	[TCP Retransmission] 44174 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519272608 TSecr=0 WS=128
653	15.002250236	10.0.2.15	8.8.8.8	TCP	74	[TCP Retransmission] 44164 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519272608 TSecr=0 WS=128
654	15.317579257	10.0.2.15	8.8.8.8	TCP	74	[TCP Retransmission] 44180 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519272803 TSecr=0 WS=128
655	16.006737180	10.0.2.15	8.8.8.8	TCP	74	[TCP Retransmission] 44164 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519273632 TSecr=0 WS=128
656	16.007112947	10.0.2.15	8.8.8.8	TCP	74	[TCP Retransmission] 44174 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519273633 TSecr=0 WS=128
657	16.341421701	10.0.2.15	8.8.8.8	TCP	74	[TCP Retransmission] 44180 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519273887 TSecr=0 WS=128
662	17.100909541	10.0.2.15	8.8.8.8	TCP	74	[TCP Retransmission] 44174 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519274655 TSecr=0 WS=128
663	17.100909529	10.0.2.15	8.8.8.8	TCP	74	[TCP Retransmission] 44164 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519274655 TSecr=0 WS=128
664	17.365202177	10.0.2.15	8.8.8.8	TCP	74	[TCP Retransmission] 44180 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519274911 TSecr=0 WS=128
665	19.150114400	10.0.2.15	8.8.8.8	TCP	74	[TCP Retransmission] 44164 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519276704 TSecr=0 WS=128

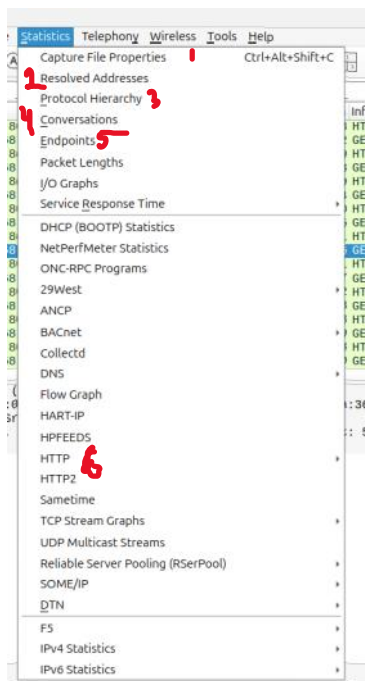
In display filter we can show different options

- IPs as wanted
- Ports
- Text or some words

No.	Time	Source	Destination	Protocol	Length	Info
2642	194.757055	10.0.0.168	13.107.5.80	HTTP	710	GET /qsm.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiiodg.&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IESS02&mark
2644	194.811822	13.107.5.80	10.0.0.168	HTTP/X..	701	HTTP/1.1 200 OK
2646	195.002742	10.0.0.168	13.107.5.80	HTTP	717	GET /qsm.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiiodg.&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IESS02&mar
2648	195.060797	13.107.5.80	10.0.0.168	HTTP/X..	702	HTTP/1.1 200 OK
2651	195.219559	10.0.0.168	13.107.5.80	HTTP	718	GET /qsm.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiiodg.e&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IESS02&ma
2653	195.287168	13.107.5.80	10.0.0.168	HTTP/X..	703	HTTP/1.1 200 OK
2655	195.424768	10.0.0.168	13.107.5.80	HTTP	719	GET /qsm.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiiodg.ex&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IESS02&m
2657	195.480048	13.107.5.80	10.0.0.168	HTTP/X..	703	HTTP/1.1 200 OK
2659	195.561991	10.0.0.168	13.107.5.80	HTTP	720	GET /qsm.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiiodg.exe&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IESS02&
2661	195.619865	13.107.5.80	10.0.0.168	HTTP/X..	703	HTTP/1.1 200 OK
2677	196.274156	10.0.0.168	103.232.55.148	HTTP	335	GET /service/.audiiodg.exe HTTP/1.1
3421	198.208642	103.232.55.148	10.0.0.168	HTTP	900	HTTP/1.1 200 OK (application/x-msdownload)

Wireshark: Statistics

Wednesday, July 9, 2025 10:52 AM



Wireshark - Capture File Properties - 2021-09-14.pcap

Details

File

Name: /home/m8a8a8y/Desktop/02_Network_Security/02_Wireshark/PCAPs/2021-09-14.pcap
Length: 2,450 kB
Hash (SHA256): 6d0c4c0b58274b25d0711f7ce29ddc8d7c0c6e594d73bddc2a1e2a90923d1c87
Hash (SHA1): c2fa875fe4618d9931fe8b05e1c10ec8e3b5a6b6
Format: Wireshark/tcpdump/... - pcap
Encapsulation: Ethernet
Snapshot length: 262144

Time

First packet: 2021-09-14 17:32:16
Last packet: 2021-09-14 17:42:01
Elapsed: 00:09:44

Capture

Hardware: Unknown
OS: Unknown
Application: Unknown

Interfaces

Interface	Dropped packets	Capture filter	Link type	Packet size limit (snaplen)
Unknown	Unknown	Unknown	Ethernet	262144 bytes

Statistics

Measurement	Captured	Displayed	Marked
Packets	3679	86 (2.3%)	—
Time span, s	584.905	276.117	—
Average pps	6.3	0.3	—
Average packet size, B	650	537	—
Bytes	2391470	46171 (1.9%)	0
Average bytes/s	4,088	167	—
Average bits/s	32 k	1,337	—

Help Refresh Edit Comments Copy To Clipboard Close

Wireshark - Resolved Addresses - /home/m8a8a8y/Desktop/02_Net...

Hosts Ports Capture File Comments

Search for entry (min 3 characters) All entries

Address	Name
ff:ff:ff:ff:ff:ff	Broadcast
01:00:5e	ICANN, IANA Department
01:00:5e:00:00:00	IPv4mcast
01:00:5e:7f:ff:fa	IPv4mcast_7f:ff:fa

Copy Save as... Close

Wireshark - Protocol Hierarchy Statistics - 2021-09-14.pcap

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	86	100.0	46171	1,337	0	0	0	86
Ethernet	100.0	86	2.6	1204	34	0	0	0	86
Internet Protocol Version 4	100.0	86	3.7	1720	49	0	0	0	86
Transmission Control Protocol	100.0	86	3.7	1720	49	0	0	0	86
Hypertext Transfer Protocol	100.0	86	2043.3	943434	27 k	45	17636	510	86
Portable Network Graphics	4.7	4	64.5	29772	862	4	29772	862	4
Online Certificate Status Protocol	7.0	6	10.6	4898	141	6	4898	141	6
Media Type	7.0	6	2099.0	969144	28 k	6	969144	28 k	6
Line-based text data	10.5	9	1127.1	520400	15 k	9	520400	15 k	9
Extensible Markup Language	15.1	13	17.7	8181	237	13	8181	237	13
Data	1.2	1	3.8	1756	50	1	1756	50	1
Compuserve GIF	2.3	2	0.8	364	10	2	364	10	2

Display filter: http

Help Protocols Copy Close

Wireshark - Conversations - 2021-09-14.pcap

Ethernet - 1	IPv4 - 10	IPv6	TCP - 24	UDP										
Address A	Address B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B
10.0.0.168	13.107.5.80	24	16 kB	18	60	40.00%	12	8 kB	12	8 kB	191.783934	71.6920	921 bits/s	878 b
10.0.0.168	23.209.125.71	2	1 kB	6	14	14.29%	1	299 bytes	1	843 bytes	31.497565	109.9813	21 bits/s	61 b
10.0.0.168	31.13.64.21	2	423 bytes	16	158	1.27%	1	361 bytes	1	62 bytes	157.902150	64.2260	44 bits/s	7 b
10.0.0.168	69.16.175.10	2	438 bytes	15	70	2.86%	1	364 bytes	1	74 bytes	154.773375	67.3575	43 bits/s	8 b
10.0.0.168	72.21.81.240	2	681 bytes	13	10	20.00%	1	336 bytes	1	345 bytes	92.170463	60.2113	44 bits/s	45 b
10.0.0.168	93.184.220.29	12	6 kB	7	52	23.08%	6	2 kB	6	4 kB	31.550561	381.1171	36 bits/s	89 b
10.0.0.168	103.232.55.148	32	18 kB	2	2,075	1.54%	16	6 kB	16	12 kB	2.434030	219.9713	210 bits/s	447 b
10.0.0.168	104.16.18.94	2	1 kB	14	23	8.70%	1	382 bytes	1	816 bytes	154.521763	67.6126	45 bits/s	96 b
10.0.0.168	104.85.5.163	2	1 kB	12	11	18.18%	1	215 bytes	1	812 bytes	68.008696	73.4689	23 bits/s	88 b
10.0.0.168	136.243.159.53	6	1 kB	19	36	16.67%	3	853 bytes	3	162 bytes	268.315691	1.4308	4,769 bits/s	905 b

Close

Wireshark - Endpoints - 2021-09-14.pcap

Ethernet - 2		IPv4 - 11		IPv6		TCP - 34		UDP											
Address	Packets	Bytes	Total Packets	Percent Filtered	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	Latitude	Longitude	AS Number	AS Organization					
10.0.0.168	86	46 kB	3,649	2.36%	43	19 kB	43	28 kB											
13.107.5.80	24	16 kB	60	40.00%	12	8 kB	12	8 kB											
23.209.125.71	2	1 kB	14	14.29%	1	843 bytes	1	299 bytes											
31.13.64.21	2	423 bytes	158	1.27%	1	62 bytes	1	361 bytes											
69.16.175.10	2	438 bytes	70	2.86%	1	74 bytes	1	364 bytes											
72.21.81.240	2	681 bytes	10	20.00%	1	345 bytes	1	336 bytes											
93.184.220.29	12	6 kB	52	23.08%	6	4 kB	6	2 kB											
103.232.55.148	32	18 kB	2,075	1.54%	16	12 kB	16	6 kB											
104.16.18.94	2	1 kB	23	8.70%	1	816 bytes	1	382 bytes											
104.85.5.163	2	1 kB	11	18.18%	1	812 bytes	1	215 bytes											
136.243.159.53	6	1 kB	36	16.67%	3	162 bytes	3	853 bytes											

Close

HPFEEDS :: 5446, Len: 602

- HTTP
 - Packet Counter
 - Requests
 - Load Distribution
 - Request Sequences
- HTTP2
- Sametime
- TCP Stream Graphs
- UDP Multicast Streams

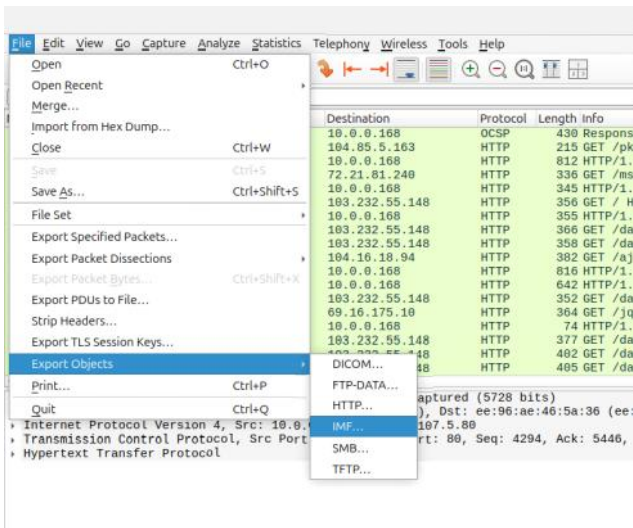
Wireshark - HTTP / Requests - 2021-09-14.pcap

Request Type	Count	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
/MEFWT2BNMEswSTAJBgUrDgMCGGUABBBQ500tx%2Fh0Zt%2Bz8SiPi7wEwVxDIQUUUBIV5uNuSg%2F6%2BkS7QYXjzCEA177eIggmWeUjG4vdGL0%3D	1	0.0000	16.67%	0.0100	278.824			
ctdl.windowsupdate.com	2	0.0000	3.92%	0.0100	31.500			
/msdownload/update/v3/static/trusted/en/pinurlstLcab7f0c731147f2df6	1	0.0000	50.00%	0.0100	92.242			
/msdownload/update/v3/static/trusted/en/disallowedcertstLcab73027e92ff72bd024	1	0.0000	50.00%	0.0100	31.500			
connect.facebook.net	1	0.0000	1.96%	0.0100	157.904			
/en_US/all.js	1	0.0000	100.00%	0.0100	157.904			
code.jquery.com	1	0.0000	1.96%	0.0100	154.776			
/query-1.10.2.min.js	1	0.0000	100.00%	0.0100	154.776			
cdnjs.cloudflare.com	1	0.0000	1.96%	0.0100	154.527			
/ajax/libs/font-awesome/3.1.0/css/font-awesome.min.css	1	0.0000	100.00%	0.0100	154.527			
api.bing.com	12	0.0000	23.53%	0.0100	191.787			
/qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IES502&market=en-US	1	0.0000	8.33%	0.0100	195.562			
/qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IES502&market=en-US	1	0.0000	8.33%	0.0100	195.425			
/qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IES502&market=en-US	1	0.0000	8.33%	0.0100	195.220			
/qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IES502&market=en-US	1	0.0000	8.33%	0.0100	195.003			
/qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IES502&market=en-US	1	0.0000	8.33%	0.0100	194.757			
/qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IES502&market=en-US	1	0.0000	8.33%	0.0100	194.589			
/qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IES502&market=en-US	1	0.0000	8.33%	0.0100	193.023			
/qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IES502&market=en-US	1	0.0000	8.33%	0.0100	192.811			
/qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IES502&market=en-US	1	0.0000	8.33%	0.0100	192.661			
/qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IES502&market=en-US	1	0.0000	8.33%	0.0100	192.259			
/qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IES502&market=en-US	1	0.0000	8.33%	0.0100	192.006			
/qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20§ionHeight=160&FORM=IES502&market=en-US	1	0.0000	8.33%	0.0100	191.787			
239.255.255.250:1900	8	0.0000	15.69%	0.0200	57.883			
*	8	0.0000	100.00%	0.0200	57.883			
136.243.159.53	3	0.0000	5.88%	0.0100	268.358			
/element/page.php?id=484	3	0.0000	100.00%	0.0100	268.358			
103.232.55.148	16	0.0001	31.37%	0.0400	154.752			
/service/audiogd.exe	1	0.0000	6.25%	0.0100	196.274			
/service/	1	0.0000	6.25%	0.0100	2.708			
/icons/blank.gif	1	0.0000	6.25%	0.0100	3.164			
/icons/back.gif	1	0.0000	6.25%	0.0100	3.165			
/favicon.ico	1	0.0000	6.25%	0.0100	3.864			
/dashboard/stylesheets/normalize.css	1	0.0000	6.25%	0.0100	154.496			
/dashboard/stylesheets/all.css	1	0.0000	6.25%	0.0100	154.752			
/dashboard/javascripts/modernizr.js	1	0.0000	6.25%	0.0100	154.785			
/dashboard/javascripts/all.js	1	0.0000	6.25%	0.0100	155.031			
/dashboard/images/kampp-logo.svg	1	0.0000	6.25%	0.0100	154.791			
/dashboard/images/social-icons.png	1	0.0000	6.25%	0.0100	158.046			

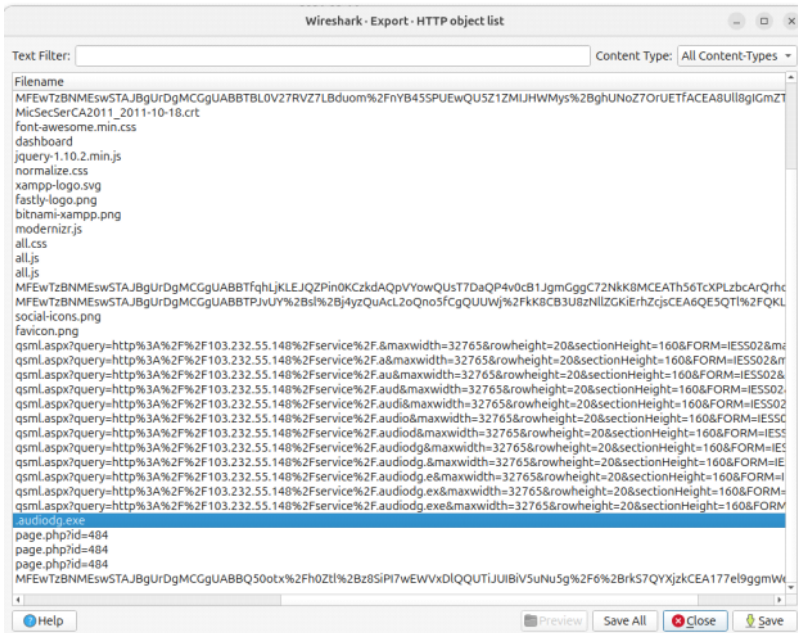
Remember it? ☺

Display filter: Enter a display filter ...

Copy Save as... Close



Form here u can install anything (take the hash or file type and see on viruse total



```
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop$ ls -la
... 01_Phishing_Analysis 02_Network_Security 03_Endpoint_Security 04_SIEM 05_Threat_Intelligence 06_Digital_Forensics 07_Incident_Response .audiodg.exe SOC
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop$ file .audiodg.exe
.audiodg.exe: PE32 executable (GUI) Intel 80386 Mono/.Net assembly, for MS Windows, 3 sections
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop$ sha256sum .audiodg.exe
f485d1a65ccf9f857baa49725d337c15e8aa34515b85c8ef59a72afad7b85249 .audiodg.exe
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop$
```

58

72

Community Score

58/72 security vendors flagged this file as malicious

ReanalyzeSimilarMore

f485d1a65ccf9f857baa49725d337c15e8aa34515b85c8ef59a72afad7b85249

Size586.50 KBLast Analysis Date1 month ago

CallerFilePathAttribu.exe

peaseassemblyruntime-modulesdirect-cpu-clock-accesschecks-user-inputspreaderdetect-debug-environmentpersistence-long-sleepsmalware

DETECTIONDETAILSRELATIONSBEHAVIORCOMMUNITY1

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat labeltrojan.msi/agenttesla

Threat categoriestrojan

Family labelsmsiagentteslaagensia

Security vendors' analysis

Do you want to automate checks?

AhnLab-V3	Malware/Win.Generic.C4631799	Alibaba	TrojanPSW:MSIL/AgentTesla.c542bc5b
AliCloud	Trojan(stealer):MSIL/AgentTesla.SZ8PHU	ALYac	Spyware.LokiBot
Arcabit	Trojan.Generic.D2CCB080	Arctic Wolf	Unsafe
Avast	Win32:MalwareX-gen [Rat]	AVG	Win32:MalwareX-gen [Rat]
Avira (no cloud)	HEUR/AGEN.1324055	BitDefender	Trojan.GenericKD.46968960

Network Security Page 21

Wireshark: Analyzing Network Traffic

Thursday, July 10, 2025 12:18 PM

Look for :

- ☐ the conversation number of packets
- ☐ See the most common IPs are talking
- ☐ See the port or the protocol being used
- ☐ Try to dig deeper example (if its http try to look for get or post request or any file upload)

File
Name: /home/m8a8a8y/Desktop/02_Network_Security/02_Wireshark/PCAPs/2023-02-03.pcap
Length: 32 MB
Hash (SHA256): c071c04241b96b91da5dad98fc1fc57da47227e2bbd8165a314c980f55a948d
Hash (SHA1): f9510a9b1e336eafcd643af65fc3a4ba059b478a
Format: Wireshark/tcpdump/... - pcap
Encapsulation: Ethernet
Snapshot length: 65535

Time
First packet: 2023-02-03 20:02:36
Last packet: 2023-02-03 22:54:43
Elapsed: 02:52:06

Capture
Hardware: Unknown
OS: Unknown
Application: Unknown

Interfaces
Interface: Unknown
Dropped packets: Unknown
Capture filter: Unknown
Link type: Ethernet
Packet size limit (snaplen): 65535 bytes

Statistics
Measurement: Captured
Packets: 55207
Time span, s: 10326.359
Average pps: 5.3
Average packet size, B: 565
Bytes: 31172531
Average bytes/s: 3,018
Average bits/s: 24 k

Displayed
55207 (100.0%)
10326.359
5.3
565
31172531 (100.0%)
3,018
24 k

Marked
—
—
—
—
0
—
—

Ethernet II	IP v4	IP v6	TCP	UDP	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
Address A	Address B												
10.0.0.149	10.0.0.6		12,040	8 MB	5	8,036	7 MB	4,004	1 MB	0.184672	10295.0052	5,115 bits/s	829 bits/s
10.0.0.149	208.187.122.14		2,860	7 MB	39	4,712	474 kB	5,094	6 MB	0.9538749	9430.4205	402 bits/s	5,392 bits/s
10.0.0.149	78.31.67.7		3,913	439 kB	49	1,947	311 kB	1,966	128 kB	1335.473325	8972.0715	277 bits/s	114 bits/s
10.0.0.149	5.75.205.43		4,127	3 MB	96	1,878	3 MB	2,249	124 kB	8328.021113	21.8323	1,005 kbps	45 kbps
10.0.0.149	74.6.143.26		2,747	2 MB	86	1,314	75 kB	1,433	2 MB	5780.995379	3027.0978	197 bits/s	5,244 bits/s
10.0.0.149	23.111.114.52		2,342	335 kB	98	1,071	235 kB	1,271	100 kB	8396.331884	1242.2290	1,514 bits/s	642 bits/s
10.0.0.149	13.107.42.14		1,449	1 MB	63	674	53 kB	775	979 kB	2537.471296	6677.8955	63 bits/s	1,173 bits/s
10.0.0.149	128.254.207.55		1,924	2 MB	25	639	35 kB	1,285	2 MB	106.401425	2.2173	124 kbps	6,605 kbps
10.0.0.149	23.58.117.164		1,020	722 kB	76	497	32 kB	523	690 kB	4253.589506	5197.2095	49 bits/s	1,062 bits/s
10.0.0.149	102.156.32.143		892	680 kB	31	471	655 kB	421	25 kB	446.974674	433.6001	12 kbps	458 bits/s
10.0.0.149	209.131.162.45		1,047	710 kB	55	470	34 kB	577	677 kB	1583.379429	7224.7145	37 bits/s	749 bits/s
10.0.0.149	23.214.54.85		877	657 kB	99	411	28 kB	466	629 kB	8515.268334	1316.3443	168 bits/s	3,823 bits/s
10.0.0.149	69.58.187.40		880	650 kB	78	400	25 kB	480	626 kB	4446.029815	4362.0639	45 bits/s	1,147 bits/s
10.0.0.149	184.86.169.24		764	550 kB	59	375	26 kB	389	524 kB	1964.661047	329.5618	624 bits/s	12 kbps
10.0.0.149	20.10.31.115		1,024	596 kB	10	350	40 kB	674	557 kB	55.493103	10216.1588	31 bits/s	436 bits/s
10.0.0.149	239.255.255.250		317	68 kB	21	317	68 kB	0	0 bytes	57.853976	10215.0031	53 bits/s	0 bits/s
10.0.0.149	23.64.146.226		524	376 kB	65	244	19 kB	280	357 kB	2730.381668	4715.2357	32 bits/s	605 bits/s
10.0.0.149	96.6.184.69		404	282 kB	95	194	14 kB	210	267 kB	8071.029433	1599.8243	70 bits/s	1,337 bits/s
10.0.0.149	142.250.68.132		295	151 kB	48	145	16 kB	150	134 kB	1331.027958	8881.4026	14 bits/s	121 bits/s
10.0.0.149	184.51.39.62		269	187 kB	30	129	9 kB	140	179 kB	445.527905	655.1113	107 bits/s	2,179 bits/s
10.0.0.149	72.163.4.185		238	78 kB	29	117	17 kB	121	61 kB	444.801479	9084.5539	15 bits/s	53 bits/s
10.0.0.149	96.7.169.183		248	180 kB	45	116	7 kB	132	173 kB	1138.770324	109.6244	535 bits/s	12 kbps
10.0.0.149	23.210.67.195		234	166 kB	80	114	7 kB	120	159 kB	4638.368628	109.6052	498 bits/s	11 kbps
10.0.0.149	104.93.55.25		225	143 kB	92	105	16 kB	120	127 kB	7115.973098	1472.1700	87 bits/s	689 bits/s
10.0.0.149	138.1.33.162		157	52 kB	50	81	12 kB	76	39 kB	1394.941651	8900.1963	10 bits/s	35 bits/s
10.0.0.149	72.247.204.96		155	93 kB	52	77	8 kB	78	85 kB	1395.397019	1059.9605	58 bits/s	643 bits/s
10.0.0.149	40.126.28.18		139	48 kB	28	76	11 kB	63	37 kB	372.306497	9720.6517	8 bits/s	30 bits/s

See what protocols are mostly used



Wireshark Challenge 1

Thursday, July 10, 2025

1:00 PM

Challenge URL

- [https://challenges.malwarecube.com/
#/c/7265ec1c-9773-4c7c-9ed4-2ea26e19f346](https://challenges.malwarecube.com/#/c/7265ec1c-9773-4c7c-9ed4-2ea26e19f346)

Intrusion Detection and Prevention Systems (IDS/IPS)

- **Intrusion Detection Systems (IDS)**
 - Passive monitoring and analysis
 - Alert generation
 - Logging
 - Out-of-band
- **Intrusion Prevention Systems (IPS)**
 - Traffic inspection
 - Blocking and dropping packets
 - Terminating connections
 - Alert generation
 - Logging
 - Inline deployment
- **Network-Based IDS/IPS (NIDS/NIPS)**
 - Operate at the network level
 - Analyze traffic flowing through network appliances
 - Alert, log, and/or block malicious traffic
- **Host-Based IDS/IPS (HIDS/HIPS)**
 - Operate at the endpoint level
 - Analyze activities on the host (file system, logins, apps)
 - Alert, log, and/or block malicious host actions

Detection Methods

- **Signature-based Detection**
 - Comparing traffic or activities against known attack patterns
 - Effective at identifying known threats
 - Cannot detect what it doesn't know exists
- **Behavior-based Detection**
 - Identifying anomalies that deviate from a pre-established baseline
 - More effective at detecting unknown threats
 - More likely to generate false positives
- **Rule-based Detection**
 - Enforcing policies to define acceptable actions or violations
 - Highly customizable to an organization's requirements
 - Requires frequent maintenance and rule updates

Known Known	Known Unknown
Unknown Known	Unknown Unknown

Introduction to Snort

Thursday, July 10, 2025 12:57 PM

What is Snort?

Snort is an open-source **network intrusion detection system (NIDS)** and **intrusion prevention system (IPS)** developed by Martin Roesch and maintained by Cisco.

Purpose:

It monitors network traffic in real time to **detect** and optionally **block** malicious activity.

Key Features:

- Packet sniffing and logging
- Attack detection using rules
- Real-time alerts
- Inline mode for blocking threats

Modes of Operation:

- **Sniffer Mode** – Displays traffic on the console
- **Packet Logger Mode** – Saves packets to disk
- **NIDS Mode** – Analyzes traffic and triggers alerts

Common Uses:

- Detecting attacks like port scans and buffer overflows
- Monitoring network traffic for anomalies
- Enhancing network security with real-time alerts

```
sudo apt install snort
```

Snort uses **Promiscuous mode**

is a network interface card (NIC) setting that allows the device to **intercept and read all traffic** on the network segment, not just the traffic addressed to it.

This is snort files having config files

```
m8a8a8y@m8a8a8y-VirtualBox:/$ cd /etc/snort
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ ls -la
total 376
drwxr-xr-x  3 root root   4096 Jul 11 15:36 .
drwxr-xr-x 141 root root 12288 Jul 11 15:36 ..
-rw-r--r--  1 root root   1281 Apr 20  2022 attribute_table.dtd
-rw-r--r--  1 root root   3757 Apr 20  2022 classification.config
-rw-r--r--  1 root root  82469 Apr 19  2024 community-sid-msg.map
-rw-r--r--  1 root root  23654 Apr 20  2022 file_magic.conf
-rw-r--r--  1 root root  33339 Apr 20  2022 gen-msg.map
-rw-r--r--  1 root root    687 Apr 20  2022 reference.config
drwxr-xr-x  2 root root   4096 Jul 11 15:36 rules
-rw-r-----  1 root snort 29773 Apr 19  2024 snort.conf
-rw-----  1 root root    803 Jul 11 15:36 snort.debian.conf
-rw-r--r--  1 root root   2335 Apr 20  2022 threshold.conf
-rw-r--r--  1 root root 160606 Apr 20  2022 unicode.map
```

The best thing is to make backup from snort.conf

```
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ sudo cp snort.conf snort.conf.bak
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ ls
attribute_table.dtd  community-sid-msg.map  gen-msg.map          rules                snort.conf.bak       threshold.conf
classification.config  file_magic.conf       reference.config     snort.conf          snort.debian.conf    unicode.map
```

Now open the configuration file

```
sudo nano snort.conf
```

```
# This file contains a sample snort configuration.
# You should take the following steps to create your own custom configuration:
#
# 1) Set the network variables.
# 2) Configure the decoder
# 3) Configure the base detection engine
# 4) Configure dynamic loaded libraries
# 5) Configure preprocessors
# 6) Configure output plugins
# 7) Customize your rule set
# 8) Customize preprocessor and decoder rule set
# 9) Customize shared object rule set
```

And can be changed

```
#####
# Step #1: Set the network variables. For more information, see README.variables
#####

# Setup the network addresses you are protecting
#
# Note to Debian users: this value is overridden when starting
# up the Snort daemon through the init.d script by the
# value of DEBIAN_SNORT_HOME_NET s defined in the
# /etc/snort/snort.debian.conf configuration file
#
ipvar HOME_NET any ← Done on installation
```

This path is used to make your own custom rules

```
#####
# Step #7: Customize your rule set
# For more information, see Snort Manual, Writing Snort Rules
#
# NOTE: All categories are enabled in this conf file
#####

# Note to Debian users: The rules preinstalled in the system
# can be *very* out of date. For more information please read
# the /usr/share/doc/snort-rules-default/README.Debian file
#
# If you install the official VRT Sourcefire rules please review this
# configuration file and re-enable (remove the comment in the first line) those
# rules files that are available in your system (in the /etc/snort/rules
# directory)

# site specific rules
include $RULE_PATH/local.rules
```

To make your own rules work make all rules in step 7 commented by adding (#) before each line

then save and test if there is no problems

```
sudo snort -T -c /etc/snort/snort.conf

+++++
Initializing rule chains...
0 Snort rules read
  0 detection rules
  0 decoder rules
  0 preprocessor rules
0 Option Chains linked into 0 Chain Headers
+++++

+-----[Rule Port Counts]-----+
|      tcp      udp      icmp      ip
|  src          0          0          0
|  dst          0          0          0
|  any          0          0          0
|  nc           0          0          0
|  s+d          0          0          0
+-----+-----+-----+-----+
```

```

Using ZLIB version: 1.3

Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.2 <Build 1>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: appid Version 1.1 <Build 5>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_SDF Version 1.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_S7COMPLUS Version 1.0 <Build 1>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_DCCRPC2 Version 1.0 <Build 3>

Total snort Fixed Memory Cost - MaxRss:48044
Snort successfully validated the configuration!
snort exiting

```

Snort modes

1. Sniffer mode

```

n8a8a8y@m8a8a8y-VirtualBox:~$ sudo snort -i enp0s3
[sudo] password for n8a8a8y:
Running in packet dump mode

--== Initializing Snort ==--
Initializing Output Plugins!
pcap DAQ configured to passive.
Acquiring network traffic from "enp0s3".
Decoding Ethernet

--== Initialization Complete ==--

o^--> Snort! <*-
      )- Version 2.9.20 GRE (Build 82)
      '--- By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
            Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
            Copyright (C) 1998-2013 Sourcefire, Inc., et al.
            Using libpcap version 1.10.4 (with TPACKET_V3)
            Using PCRE version: 8.39 2016-06-14
            Using ZLIB version: 1.3

Commencing packet processing (pid=4023)
07/11-16:10:26.308329 10.0.2.15:54339 -> 10.0.2.3:53
UDP TTL:64 TOS:0x0 ID:44925 IpLen:20 DgmLen:67
Len: 39
=====

07/11-16:10:26.308981 10.0.2.15:55396 -> 10.0.2.3:53
UDP TTL:64 TOS:0x0 ID:27052 IpLen:20 DgmLen:67
Len: 39
=====

WARNING: No preprocessors configured for policy 0.
07/11-16:10:26.327380 10.0.2.3:53 -> 10.0.2.15:55396
UDP TTL:64 TOS:0x0 ID:3464 IpLen:20 DgmLen:95
Len: 67

```

```

n8a8a8y@m8a8a8y-VirtualBox:~$ curl google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
n8a8a8y@m8a8a8y-VirtualBox:~$

```

Adding -e will give more info

```

n8a8a8y@m8a8a8y-VirtualBox:~$ sudo snort -i enp0s3 -e
Running in packet dump mode

--== Initializing Snort ==--
Initializing Output Plugins!
pcap DAQ configured to passive.
Acquiring network traffic from "enp0s3".
Decoding Ethernet

--== Initialization Complete ==--

o^--> Snort! <*-
      )- Version 2.9.20 GRE (Build 82)
      '--- By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
            Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
            Copyright (C) 1998-2013 Sourcefire, Inc., et al.
            Using libpcap version 1.10.4 (with TPACKET_V3)
            Using PCRE version: 8.39 2016-06-14
            Using ZLIB version: 1.3

Commencing packet processing (pid=4059)
07/11-16:14:26.525081 08:00:27:A4:86:ED -> 52:55:0A:00:02:02 type:0x800 len:0x4A
10.0.2.15:48280 -> 142.251.37.46:80 TCP TTL:64 TOS:0x0 ID:53627 IpLen:20 DgmLen:60 DF
*****S* Seq: 0xD6C7BE0A Ack: 0x0 Win: 0xFAF0 TcpLen: 40
TCP Options (5) => MSS: 1460 SackOK TS: 3191672462 0 NOP WS: 7
=====

WARNING: No preprocessors configured for policy 0.
07/11-16:14:26.585807 52:55:0A:00:02:02 -> 08:00:27:A4:86:ED type:0x800 len:0x3C
142.251.37.46:80 -> 10.0.2.15:48280 TCP TTL:64 TOS:0x0 ID:3489 IpLen:20 DgmLen:44
***A**S* Seq: 0xA4EAC01 Ack: 0xD6C7BE0B Win: 0xFFFF TcpLen: 24
TCP Options (1) => MSS: 1460
=====

```

Adding the -d will give you the ASCII

```
m8a8a8y@m8a8a8y-VirtualBox: $ sudo snort -i enp0s3 -d
Running in packet dump mode

--== Initializing Snort ==--
Initializing Output Plugins!
pcap DAQ configured to passive.
Acquiring network traffic from "enp0s3".
Decoding Ethernet

--== Initialization Complete ==--

o" )-  -> Snort! <-
  ""   Version 2.9.20 GRE (Build 82)
  ""   By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
  ""   Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
  ""   Copyright (C) 1998-2013 Sourcefire, Inc., et al.
  ""   Using libpcap version 1.10.4 (with TPACKET_V3)
  ""   Using PCRE version: 8.39 2016-06-14
  ""   Using ZLIB version: 1.3

Commencing packet processing (pid=4001)
07/11-16:17:26.086931 10.0.2.15:57363 -> 10.0.2.3:53
UDP TTL:64 TOS:0x0 ID:7656 IpLen:20 DgmLen:67
Len: 39
17 DC 01 00 00 01 00 00 00 00 01 06 67 6F 6F .....goo
67 6C 65 03 63 6F 6D 00 00 1C 00 01 00 00 29 05 gle.com.....).
C0 00 00 00 00 00 00 .....
=====

WARNING: No preprocessors configured for policy 0.
07/11-16:17:26.105101 10.0.2.3:53 -> 10.0.2.15:57363
UDP TTL:64 TOS:0x0 ID:3500 IpLen:20 DgmLen:95
Len: 67
17 DC 81 80 00 01 00 01 00 00 01 06 67 6F 6F .....goo
```

2. Packet logger mode

```
m8a8a8y@m8a8a8y-VirtualBox:~/log$ sudo snort -i enp0s3 -l .
Running in packet logging mode

--== Initializing Snort ==--
Initializing Output Plugins!
Log directory = .
pcap DAQ configured to passive.
Acquiring network traffic from "enp0s3".
Decoding Ethernet

--== Initialization Complete ==--

o" )-  -> Snort! <-
  ""   Version 2.9.20 GRE (Build 82)
  ""   By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
  ""   Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
  ""   Copyright (C) 1998-2013 Sourcefire, Inc., et al.
  ""   Using libpcap version 1.10.4 (with TPACKET_V3)
  ""   Using PCRE version: 8.39 2016-06-14
  ""   Using ZLIB version: 1.3

Commencing packet processing (pid=4152)
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
^C*** Caught Int-Signal
WARNING: No preprocessors configured for policy 0.
=====
```

```
m8a8a8y@m8a8a8y-VirtualBox:~/log$ ls
snort.log.1752240294
m8a8a8y@m8a8a8y-VirtualBox:~/log$ sudo snort -r snort.log.1752240294
Running in packet dump mode

--== Initializing Snort ==--
Initializing Output Plugins!
pcap DAQ configured to read-file.
Acquiring network traffic from "snort.log.1752240294".

--== Initialization Complete ==--

o" )-  -> Snort! <-
  ""   Version 2.9.20 GRE (Build 82)
  ""   By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
  ""   Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
  ""   Copyright (C) 1998-2013 Sourcefire, Inc., et al.
  ""   Using libpcap version 1.10.4 (with TPACKET_V3)
  ""   Using PCRE version: 8.39 2016-06-14
  ""   Using ZLIB version: 1.3

Commencing packet processing (pid=4175)
07/11-16:24:56.527763 10.0.2.15:60468 -> 10.0.2.3:53
UDP TTL:64 TOS:0x0 ID:25725 IpLen:20 DgmLen:67
Len: 39
=====
07/11-16:24:56.528731 10.0.2.15:60468 -> 10.0.2.3:53
```

```

m8a8a8y@m8a8a8y-VirtualBox:~/logs$ ls
snort.log.1752240294
m8a8a8y@m8a8a8y-VirtualBox:~/logs$ sudo snort -r snort.log.1752240294
Running in packet dump mode

--== Initializing Snort ==--
Initializing Output Plugins!
pcap DAQ configured to read-file.
Acquiring network traffic from "snort.log.1752240294".

--== Initialization Complete ==--

o''-~
''''
-*)<*-
Version 2.9.20 GRE (Build 82)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.10.4 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.3

Commencing packet processing (pid=4175)
07/11-16:24:56.527763 10.0.2.15:60468 -> 10.0.2.3:53
UDP TTL:64 TOS:0x0 ID:25725 IpLen:20 DgmLen:67
Len: 39
=====
07/11-16:24:56.528721 10.0.2.15:52357 -> 10.0.2.3:53
UDP TTL:64 TOS:0x0 ID:4522 IpLen:20 DgmLen:67
Len: 39
=====

```

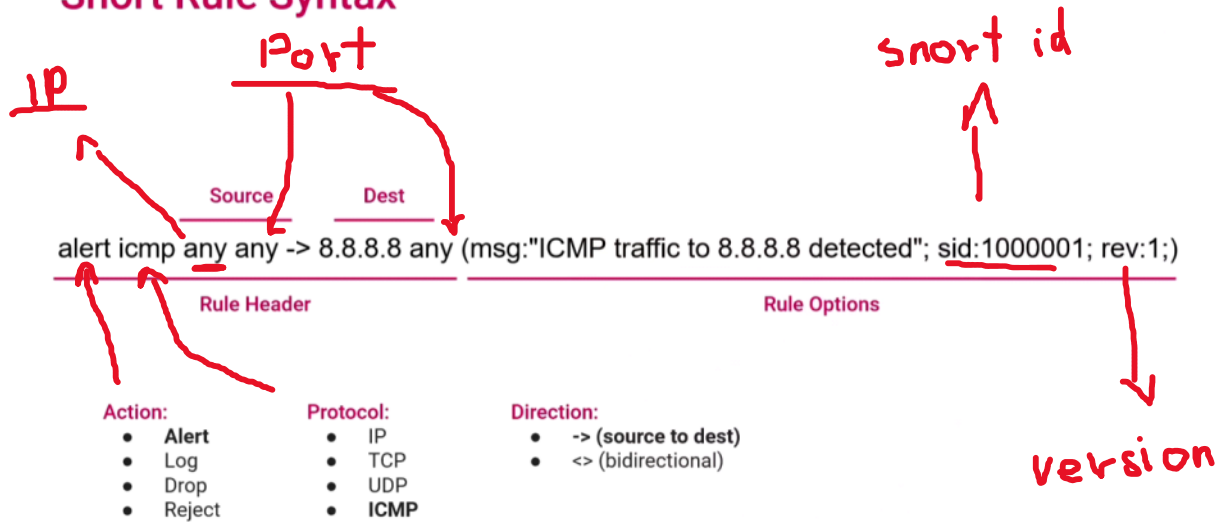
Also tcpdump can use this file

```

m8a8a8y@m8a8a8y-VirtualBox:~/logs$ sudo tcpdump -r snort.log.1752240294
reading from file snort.log.1752240294, link-type EN10MB (Ethernet), snapshot length 1514
16:24:56.527763 IP m8a8a8y-VirtualBox.60468 > 10.0.2.3.domain: 38191+ [1au] A? google.com. (39)
16:24:56.528721 IP m8a8a8y-VirtualBox.52357 > 10.0.2.3.domain: 13458+ [1au] AAAA? google.com. (39)
16:24:56.543472 IP 10.0.2.3.domain > m8a8a8y-VirtualBox.52357: 13458 1/0/1 AAAA 2a00:1450:4006:810::200e (67)
16:24:56.543477 IP 10.0.2.3.domain > m8a8a8y-VirtualBox.60468: 38191 1/0/1 A 142.251.37.46 (55)
16:24:56.545836 IP m8a8a8y-VirtualBox.60556 > mrs09s13-in-f14.1e100.net.http: Flags [S], seq 331796290, win 64240, options [mss 1460,sackOK,TS val 3192302482 ecr 0,nop,wscale 7], length 0
16:24:56.610211 IP mrs09s13-in-f14.1e100.net.http > m8a8a8y-VirtualBox.60556: Flags [S.], seq 182016001, ack 1331796291, win 65535, options [mss 1460], length 0
16:24:56.610336 IP m8a8a8y-VirtualBox.60556 > mrs09s13-in-f14.1e100.net.http: Flags [.], ack 182016001, win 64240, length 0
16:24:56.610873 IP m8a8a8y-VirtualBox.60556 > mrs09s13-in-f14.1e100.net.http: Flags [P.], seq 1:74, ack 1, win 64240, length 73: HTTP: GET / HTTP/1.1
16:24:56.611646 IP mrs09s13-in-f14.1e100.net.http > m8a8a8y-VirtualBox.60556: Flags [.], ack 74, win 65535, length 0
16:24:56.764251 IP mrs09s13-in-f14.1e100.net.http > m8a8a8y-VirtualBox.60556: Flags [P.], seq 1:774, ack 74, win 65535, length 773: HTTP: HTTP/1.1 301 Moved Permanently
16:24:56.764409 IP m8a8a8y-VirtualBox.60556 > mrs09s13-in-f14.1e100.net.http: Flags [.], ack 774, win 63467, length 0
16:24:56.765001 IP m8a8a8y-VirtualBox.60556 > mrs09s13-in-f14.1e100.net.http: Flags [F.], seq 774, ack 774, win 63467, length 0
16:24:56.765328 IP mrs09s13-in-f14.1e100.net.http > m8a8a8y-VirtualBox.60556: Flags [.], ack 774, win 65535, length 0
16:24:56.827739 IP mrs09s13-in-f14.1e100.net.http > m8a8a8y-VirtualBox.60556: Flags [F.], seq 774, ack 75, win 65535, length 0
16:24:56.827771 IP m8a8a8y-VirtualBox.60556 > mrs09s13-in-f14.1e100.net.http: Flags [.], ack 75, win 63467, length 0
16:25:02.000548 ARP, Request who-has _gateway tell m8a8a8y-VirtualBox, length 28

```

Snort Rule Syntax



```
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ ls
attribute_table.dtd  community-sid-msg.map  gen-msg.map  rules  snort.conf.bak  threshold.conf
classification.config  file_magic.conf  reference.config  snort.conf  snort.debian.conf  unicode.map
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ sudo nano /rules/local.rules
```

```
GNU nano 7.2 local.rules
## $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures.  Put your local
# additions here.
```

```

GNU nano 7.2                               rules/local.rules *
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures.  Put your local
# additions here.
alert icmp any any -> 8.8.8.8 any (msg: "ICMP traffic to 8.8.8.8 detected"; sid: 1000001; rev:1;)
```

-A alert-mode

Alert using the specified **alert-mode**. Valid alert modes include **fast**, **full**, **none**, and **unsock**. **Fast** writes alerts to the default "alert" file in a single-line, syslog style alert message. **Full** writes the alert to the "alert" file with the full decoded header as well as the alert message. **None** turns off alerting. **Unsock** is an experimental mode that sends the alert information out over a UNIX socket to another process that attaches to that socket.

```
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ sudo snort -A console -l /var/log/snort -i enp0s3 -c /etc/snort/snort.conf -q
```

Display alerts on the command

interface

Config file of snort

Quite mode no details
If not used its OK

If we ping 8.8.4.4

```

m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ sudo snort -A console -l /var/log/snort/ -i enp0s3 -c /etc/snort/snort.conf -q
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ ping 8.8.4.4
PING 8.8.4.4 (8.8.4.4) 56(84) bytes of data.
64 bytes from 8.8.4.4: icmp_seq=1 ttl=255 time=65.2 ms
64 bytes from 8.8.4.4: icmp_seq=2 ttl=255 time=66.4 ms
64 bytes from 8.8.4.4: icmp_seq=3 ttl=255 time=69.5 ms
64 bytes from 8.8.4.4: icmp_seq=4 ttl=255 time=65.4 ms
64 bytes from 8.8.4.4: icmp_seq=5 ttl=255 time=66.8 ms
^C
--- 8.8.4.4 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4009ms
rtt min/avg/max/mdev = 65.236/66.659/69.466/1.520 ms
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$
```

And if 8.8.8.8

```

m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ sudo snort -A console -l /var/log/snort/ -i enp0s3 -c /etc/snort/snort.conf -q
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=67.9 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=65.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=70.5 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=255 time=65.1 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=255 time=65.4 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=255 time=66.8 ms
^C
--- 8.8.8.8 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5010ms
rtt min/avg/max/mdev = 65.138/66.854/70.455/1.876 ms
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$
```

<https://snort.org/downloads#rule-downloads> : has many ready rules can be used

```
tcm@SOC101-ubuntu:/var/log/snort$ ls
snort.alert.fast  snort.log.1717962723  snort.log.1717963124
tcm@SOC101-ubuntu:/var/log/snort$ sudo wireshark
```

Apply a display filter ... «Ctrl+F»									
No.	Time	Source	Source Port	Destination	Dest Port	Protocol	Length	Info	
1	2024-06-09 19:58:54.422137	192.168.1.4	2125	93.184.215.14	4444	TCP	54	2125 → 4444 [SYN] Seq=0 Win=512 Len=0	

U can use snorpy -> <https://snorpy.cyb3rs3c.net/>
To make rules

SNORPY

A Web Based Snort Rule Creator / Maker for Building Simple Snort Rules

alert

tcp

any

any

»

8.8.8.8

4444

1000001

1

Rule Message (\ Escape special characters)

Class-Type

Priority

gid

TCP

HTTP REQUEST METHOD HTTP RESPONSE CODE

ACK ☐ SYN ☐ PSH ☐ RST ☐ FIN ☐ URG ☐ * ☐ *

DIRECTION TCP STATE

Data Size

Reference

Threshold Tracking Type TRK BY Count # Seconds

Add Content Match

Add Regex Match

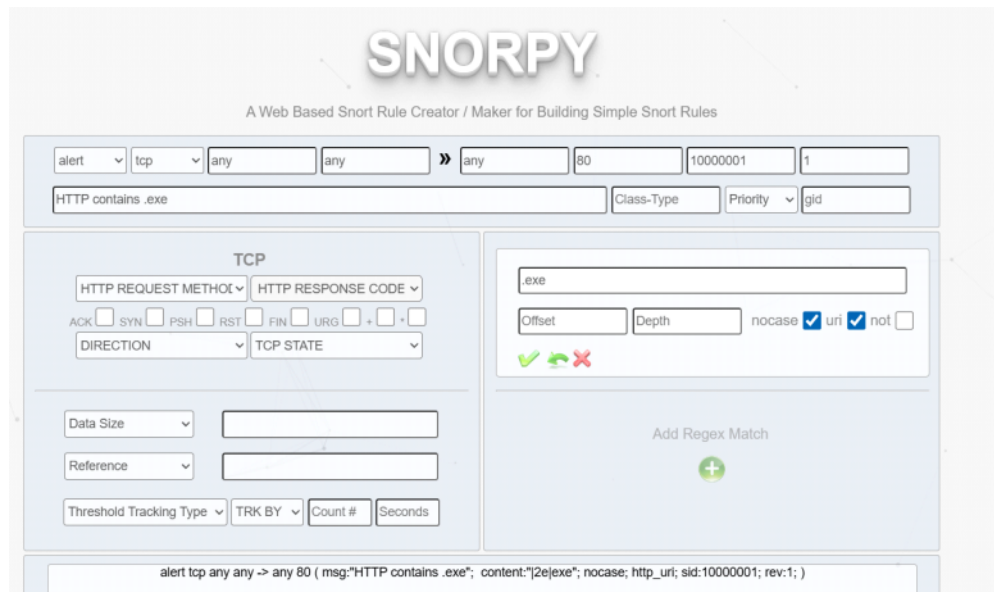
alert tcp any any -> 8.8.8.8 4444 (sid:1000001; rev:1;)

Snort: Intrusion Detection and Prevention

Thursday, July 10, 2025 12:57 PM

We are going to make an alert in exe file on a pcap

```
m8a8a8y@m8a8a8y-VirtualBox:~/Desktop/02_Network_Security/03_Snort/PCAPs$ sudo wireshark 1.pcap
[sudo] password for m8a8a8y:
** (wireshark:2538) 20:33:17.384310 [GUI WARNING] -- QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-root'
```



```
m8a8a8y@m8a8a8y-VirtualBox:~/Desktop/02_Network_Security/03_Snort/PCAPs$ sudo nano /etc/snort/rules/local.rules
```

```
GNU nano 7.2 /etc/snort/rules/local.rules *
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures. Put your local
# additions here.
alert tcp any any -> any 80 ( msg:"HTTP contains .exe"; content:"|2e|exe"; nocase; http_uri; sid:10000001; rev:1; )
```

```
m8a8a8y@m8a8a8y-VirtualBox:~/Desktop/02_Network_Security/03_Snort/PCAPs$ sudo snort -c /etc/snort/snort.conf -q -r 1.pcap -A console
09/14-17:35:31.840413 [**] [1:10000001:1] HTTP contains .exe [**] [Priority: 0] {TCP} 10.0.0.168:49722 -> 13.107.5.80:80
09/14-17:35:32.552578 [**] [1:10000001:1] HTTP contains .exe [**] [Priority: 0] {TCP} 10.0.0.168:49724 -> 103.232.55.148:80
```

```
m8a8a8y@m8a8a8y-VirtualBox:~/Desktop/02_Network_Security/03_Snort/PCAPs$ sudo snort -q -d -r /var/log/snort/snort.log.1752342302
09/14-17:35:31.840413 10.0.0.168:49722 -> 13.107.5.80:80
TCP TTL:128 TOS:0x0 ID:502 IpLen:20 DgmLen:706 DF
***AP*** Seq: 0x53A34ABB Ack: 0xD074CC7D Win: 0x7FAE TcpLen: 20
47 45 54 20 2F 71 73 6D 6C 2E 61 73 70 78 3F 71 GET /qsm1.aspx?q
75 65 72 79 3D 68 74 74 70 25 33 41 25 32 46 25 uery=http%3A%2F%
32 46 31 30 33 2E 32 33 32 2E 35 35 2E 31 34 38 2F103.232.55.148
25 32 46 73 65 72 76 69 63 65 25 32 46 2E 61 75 %2Fservice%2F.au
64 69 6F 64 67 2F 65 78 65 26 6D 61 78 77 69 64 diogn.exe&maxwid
```

```
47 45 54 20 2F 71 73 6D 6C 2E 61 73 70 78 3F 71 GET /qsm.aspx?q
75 65 72 79 3D 68 74 74 70 25 33 41 25 32 46 25 uery=http%3A%2F%
32 46 31 30 33 2E 32 33 32 2E 35 35 2E 31 34 38 2F103.232.55.148
25 32 46 73 65 72 76 69 63 65 25 32 46 2E 61 75 %2Fservice%2F.au
64 69 6F 64 67 2E 65 78 65 26 6D 61 78 77 69 64 diodg.exe&maxwid
74 68 3D 33 32 37 36 35 26 72 6F 77 68 65 69 67 th=32765&rowheig
68 74 3D 32 30 26 73 65 63 74 69 6F 6E 48 65 69 ht=20&sectionHei
67 68 74 3D 31 36 30 26 46 4F 52 4D 3D 49 45 53 ght=160&FORM=IES
53 30 32 26 6D 61 72 68 65 74 3D 65 6E 2D 55 53 S02&market=en-US
20 48 54 54 50 2F 31 2E 31 0D 0A 41 63 63 65 70 HTTP/1.1..Accep
74 3A 20 2A 2F 2A 0D 0A 41 63 63 65 70 74 2D 4C t: /*..Accept-L
61 6E 67 75 61 67 65 3A 20 65 6E 2D 55 53 0D 0A nguage: en-US..
55 73 65 72 2D 41 67 65 6E 74 3A 20 4D 6F 7A 69 User-Agent: Mozi
6C 6C 61 2F 35 2E 30 28 57 69 6E 64 6F 77 73 lla/5.0 (Windows
20 4E 54 20 31 30 2E 30 38 20 57 4F 57 36 34 38 NT 10.0; WOW64;
20 54 72 69 64 65 6E 74 2F 37 2E 30 38 20 72 76 Trident/7.0; rv
3A 31 31 2E 30 29 20 6C 69 68 65 20 47 65 63 6B :11.0) like Geck
6F 0D 0A 41 63 63 65 70 74 2D 45 6E 63 6F 64 69 o..Accept-Encodi
6E 67 3A 20 67 7A 69 70 2C 20 64 65 66 6C 61 74 ng: gzip, deflat
65 0D 0A 48 6F 73 74 3A 20 61 70 69 2E 62 69 6E e..Host: api.bin
67 2E 63 6F 6D 0D 0A 43 6F 6E 6E 65 63 74 69 6F g.com..Connectio
6E 3A 20 48 65 65 70 2D 41 6C 69 76 65 0D 0A 43 n: Keep-Alive..C
6F 6F 68 69 65 3A 20 4D 55 49 44 3D 33 42 31 32 ookie: MUID=3B12
36 41 38 46 35 38 41 45 36 45 44 37 33 39 38 41 6A8F58AE6ED7398A
37 41 33 44 35 39 30 44 36 46 34 35 38 20 5F 45 7A3D590D6F45; _E
44 47 45 5F 53 3D 46 3D 31 26 53 49 44 3D 33 38 DGE S=F=1&SID=38
```

quite

Not files are being sent with .exe some are transmitted as
And we can use it to trigger alert

And here we want the response header on port 80

SNORPY

A Web Based Snort Rule Creator / Maker for Building Simple Snort Rules

alert
tcp
any
80
»
any
any
10000001
?

HTTP contains .exe
Class-Type
Priority
gid

TCP

HTTP REQUEST METHOD
HTTP RESPONSE CODE

ACK ☐
SYN ☐
PSH ☐
RST ☐
FIN ☐
URG ☐
+
+

DIRECTION
TCP STATE

Data Size

Reference

Threshold Tracking Type
TRK BY
Count #
Seconds

Content-Type: application/x-msdownload

Offset
Depth
nocase ☒
uri ☒
not ☐

✓
↔
✗

Add Regex Match
+

```

alert tcp any 80 -> any any ( msg:"HTTP contains .exe"; content:"[2e]exe"; nocase; http_uri; content:"Content[2d]Type[3a
20]application[2f]x[2d]msdownload"; nocase; http_uri; sid:10000001; rev:1; )

```

Try to make the rest alone

Aslo we can use the files siganture



Snort Challenge 1

Thursday, July 10, 2025

12:57 PM

<https://challenges.malwarecube.com/#/c/fec27521-e154-4637-a1bd-38777d85c974>

Additional Practice

Thursday, July 10, 2025 12:58 PM

References

- <https://www.malware-traffic-analysis.net/>
- <https://chrissanders.org/packet-captures/>
- <https://github.com/chrissanders/packets>
- <https://wiki.wireshark.org/SampleCaptures>
- <https://www.netresec.com/?page=PcapFiles>
- <https://github.com/zeek/zeek/tree/master/testing/btest/Traces>
- <https://docs.securityonion.net/en/2.4/pcaps.html>